

NATTAVAARA, Sweden

WMO: 021610

Lat: 66.752N

Lon: 20.928E

Elev: 1073

StdP: 14.14

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-22.2	-35.7	0.7	-28.2	-29.0	1.1	-21.8	14.2	23.7	12.8	23.5	0.6	290	0.695

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	75.3	59.6	71.7	58.0	68.2	56.9	62.8	71.5	60.4	67.9	58.4	65.3	5.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.4	78.7	65.2	57.3	72.9	62.8	55.5	68.2	60.9	28.9	71.6	27.1	68.1	25.8	65.5	70.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.0	10.3	-35.0	80.1	5.7	6.1	-39.1	84.4	-42.4	88.0	-45.6	91.4	-49.7	95.8		
(5)			-33.4	65.9	5.8	2.8	-37.5	67.9	-40.9	69.6	-44.2	71.2	-48.4	73.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.1	7.7	8.7	17.6	29.0	41.6	51.8	56.9	53.0	43.6	30.3	19.0	12.8		
	DBStd	19.71	13.78	13.80	11.07	7.82	7.29	6.61	5.07	5.88	6.08	8.51	11.06	12.76		
	HDD50	7380	1313	1156	1004	630	279	55	36	206	612	929	1155			
	HDD65	12375	1778	1576	1469	1080	725	399	256	374	642	1077	1379	1620		
	CDD50	488	0	0	0	0	18	109	219	128	14	0	0	0		
	CDD65	9	0	0	0	0	0	3	5	1	0	0	0	0		
	CDH74	171	0	0	0	0	9	52	92	18	0	0	0	0		
	CDH80	27	0	0	0	0	0	10	17	0	0	0	0	0		
	WSAvg	4.2	3.0	3.4	4.8	4.6	5.2	5.6	4.8	4.2	4.3	4.0	3.1	3.1		
	PrecAvg	23.5	1.7	1.2	1.2	1.1	1.7	2.6	3.5	2.9	2.1	1.9	1.9	1.7		
(15) Precipitation	PrecMax	32.4	3.2	2.3	2.4	2.3	3.4	4.2	4.9	5.4	7.7	3.6	4.0	5.7		
	PrecMin	16.1	0.2	0.2	0.4	0.3	0.4	0.6	1.2	1.1	0.6	0.6	0.6	0.4		
	PrecStd	4.2	0.7	0.6	0.6	0.6	0.8	1.0	1.1	1.3	1.5	0.8	1.0	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	41.0	44.6	57.2	74.6	80.3	81.9	76.6	64.2	50.8	42.6	38.9		
		MCWB	36.5	36.3	37.8	44.0	54.3	62.5	64.2	62.3	54.6	45.6	39.5	35.7		
	2%	DB	33.0	36.4	40.3	50.0	68.2	73.3	76.1	72.2	60.3	47.0	38.6	34.1		
		MCWB	31.3	32.8	34.7	40.5	53.0	56.0	60.8	60.4	52.7	44.0	37.0	32.2		
	5%	DB	29.7	32.1	37.2	45.9	61.3	69.1	72.2	68.3	57.0	44.2	34.7	31.9		
		MCWB	28.9	30.0	32.8	38.4	49.4	55.0	59.4	58.8	51.4	41.7	33.7	31.1		
	10%	DB	26.0	28.6	34.0	42.2	55.6	65.3	68.5	64.4	54.0	41.4	32.4	29.9		
		MCWB	25.3	27.6	30.7	36.1	45.9	53.4	57.9	56.6	49.5	39.5	31.9	29.1		
	0.4%	WB	36.7	36.7	37.8	44.6	57.1	63.9	66.6	64.8	56.6	47.1	40.3	36.0		
		MCDB	39.5	40.5	43.6	53.1	70.2	75.3	73.3	72.6	61.0	49.2	41.3	38.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.6	33.3	35.2	41.5	54.0	59.5	63.6	62.0	54.3	44.5	36.9	32.5		
		MCDB	32.5	35.8	39.4	49.1	66.2	68.4	72.8	69.3	59.1	47.0	38.7	33.5		
	5%	WB	29.0	30.1	33.3	38.9	50.3	56.7	61.1	59.7	52.1	42.0	33.9	31.1		
		MCDB	29.9	31.6	36.6	45.3	60.7	65.2	68.9	66.5	55.5	44.1	34.6	31.8		
	10%	WB	25.3	27.5	31.0	36.4	47.1	54.4	59.1	57.6	50.4	39.7	32.1	29.0		
		MCDB	25.9	28.8	33.7	41.7	54.5	63.2	66.2	62.9	53.4	41.2	32.4	29.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	19.0	22.0	19.3	17.6	17.2	17.9	17.4	15.7	12.1	13.2	16.0		
		MCDBR	15.5	16.9	21.6	23.5	29.2	24.7	25.5	24.6	20.8	14.8	11.2	13.8		
		MCWBR	14.8	15.3	16.9	15.1	15.5	11.3	12.5	13.6	13.7	11.8	10.3	13.2		
		MCDBR	14.7	16.3	19.8	21.2	27.2	21.2	21.1	21.3	17.4	13.8	10.4	13.7		
	5% WB	MCWBR	14.1	14.8	16.0	13.6	15.1	11.0	12.0	12.5	12.1	11.3	9.8	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.182	0.247	0.282	0.312	0.330	0.333	0.342	0.327	0.297	0.273	0.341	N/A		
	taud		1.938	2.146	2.226	2.280	2.467	2.496	2.483	2.510	2.572	2.395	2.772	N/A		
	Ebn at Noon		68	191	246	268	277	278	271	264	248	187	77	0		
	Edh at Noon		4	16	25	31	29	29	29	25	18	13	5	0		
(44) All-Sky Solar Radiation	RadAvg		22	193	670	1270	1570	1632	1521	1120	626	223	44	0		
	RadStd		2	24	45	58	152	127	148	99	89	36	5	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page